

time-dependent Hamiltonian vector field of  $\tau^* \frac{\partial S_t}{\partial t}$  starting at  $d_x S_0$ . Equation (2.24) is then a plain consequence of the chain rule. Since

$$\forall f, g \in \mathcal{C}^\infty(T^*M), \quad d\{f, g\}_\omega = [df, dg], \quad (2.25)$$

equation (2.23) is obtained by usual extension from smooth functions to exact 1-forms, and then from closed forms to generic 1-forms using the Leibniz rule.  $\square$

Let us illustrate Lemma 2.17. The algebroid bracket  $[\cdot, \cdot]$  allows us to obtain iterated derivations of the Hamilton-Jacobi equation (2.21). We spell out the order 2 :

$$\frac{\partial^2 \zeta_t}{\partial t^2} = (\zeta_t)^* [\alpha^* \theta, \tau^* \zeta_t \alpha^* \theta], \quad (2.26)$$

After applying Lemma 2.17 a second time, we obtain an order 3 derivation:

$$\frac{\partial^3 \zeta_t}{\partial t^3} = (\zeta_t)^* ([\alpha^* \theta, \tau^* (\zeta_t)^* [\alpha^* \theta, \tau^* (\zeta_t)^* \alpha^* \theta]] + [[\alpha^* \theta, \tau^* (\zeta_t)^* \alpha^* \theta], \tau^* (\zeta_t)^* \alpha^* \theta]). \quad (2.27)$$

### 3 A Pre-Lie approach to Hamiltonian Poisson integrators

Since  $\mathcal{G} \subset T^*M$ , the Lagrangian bisections of  $\mathcal{G}$  that are close to the identity section are described by graphs of time-dependent closed 1-forms. We can approximate them by using the notion of jets we developed in the previous section. In order to approximate a given Lagrangian bisection by 1-forms, we now introduce an appropriate space  $J_\xi^\infty$  and a pre-Lie algebra structure on it. We then explain how this pre-Lie algebra encodes expansions of solutions of the Hamilton-Jacobi equation (2.21) through the introduction of Butcher series. In Section 3.3, we exhibit some algebraic simplifications arising in this expansion if the initial condition is chosen to be zero.

#### 3.1 Pre-Lie formalism for Hamilton-Jacobi flows

In full generality, for  $\xi \in \Omega^1(\mathcal{G})$ , we are interested in the expansion of the general Hamilton-Jacobi flow

$$\frac{\partial \zeta}{\partial t} = \zeta_t^* \xi, \quad \zeta_0 \in \Omega_0^1(M). \quad (3.1)$$

We consider the infinite dimensional real vector subspace  $J_\xi^\infty \subset \Omega^1(\mathcal{G})$  spanned by  $\xi$  and stable w.r.t. the following maps: for any  $f, g \in J_\xi^\infty$ ,

$$\eta(f, g): \begin{array}{ccc} \Omega_0^1(M) & \rightarrow & \Omega_0^1(M) \\ \zeta & \mapsto & \zeta^* [f, \tau^* \zeta^* g] \end{array} . \quad (3.2)$$

The space  $J_\xi^\infty$  is naturally equipped with the product  $\triangleright$

$$h \triangleright \xi(\zeta) = \zeta^* [\xi, \tau^* \zeta^* g], \quad h \triangleright \eta(f, g) = \eta(h \triangleright f, g) + \eta(f, h \triangleright g). \quad (3.3)$$

This yields a non-associative non-commutative magmatic structure on  $(J_\xi^\infty, \triangleright)$ . Note that  $(J_\xi^\infty, \triangleright)$  is not a Lie algebra as  $\triangleright$  is not antisymmetric.

**Remark 3.1** ( $J_\xi^\infty$  and jet spaces). *The space  $J_\xi^\infty$  is analogous to the infinite jet space [1, 41, 37] on  $\mathfrak{X}(M)$  used for the analysis of Runge-Kutta and Lie-group methods. In [31, 27], the accuracy of numerical integrators is studied via the use of Taylor expansions of ODE flows. Given a vector*